ANALYTICS FOR BUSINESS LAB

acceleration
A WPP Company

GROUP M - MARKETING MIX MODELLING

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Methodology

I. Setting of objectives

II. Identification of preliminary data and data owners

III. Preliminary Data Analysis

iV. Model selection, model fit, and hyperparameter optimization

VI. Model evaluation

VII. Interpretation of the model and results analysis

I. OBJECTIVES AND GOALS

- Understanding the dynamics of business and what drives sales.
- Understanding the effects of each factor(promo, media, etc) on sales and how they change on a year-on-year basis.
- Deep dive into media investments and contributions.
- Total effectiveness, total Return on Investments and the optimizing the media mix.

II. IDENTIFICATION OF PRELIMINARY DATA AND DATA OWNERS

The background provided for this MMM project:

- A comparatively young beauty brand that has gained prominence in the market over the last five years.
- Their primary goal is to increase their market share, by increasing the recognition of their product in the market. To achieve this, they want to optimize their media and promotional investments.

III. PRELIMINARY DATA ANALYSIS

For this step we carried out the following things:

1. **<u>Data cleaning and assessment</u>:**

- Checked for missing values and replaced them with 0.
- The data was not arranged according to the date. Aligned the data to start from the first week of 2020.
- The data types were different for different categories. Converted all the categories into numbers, in order to make the data analysis more consistent.
- Checked for duplicate values in the columns.

2. **<u>Data Transformation</u>:**

- First of all, an Inflation Index column was added and the values were generated using 2020 as the base year. The formula used for generating the values was

  *Adjusted Value = Original Value * (Consumer Price Index in Target Year)/(Consumer Price Index in Base Year)*

  This adjustment was made for every category that represented a currency value.

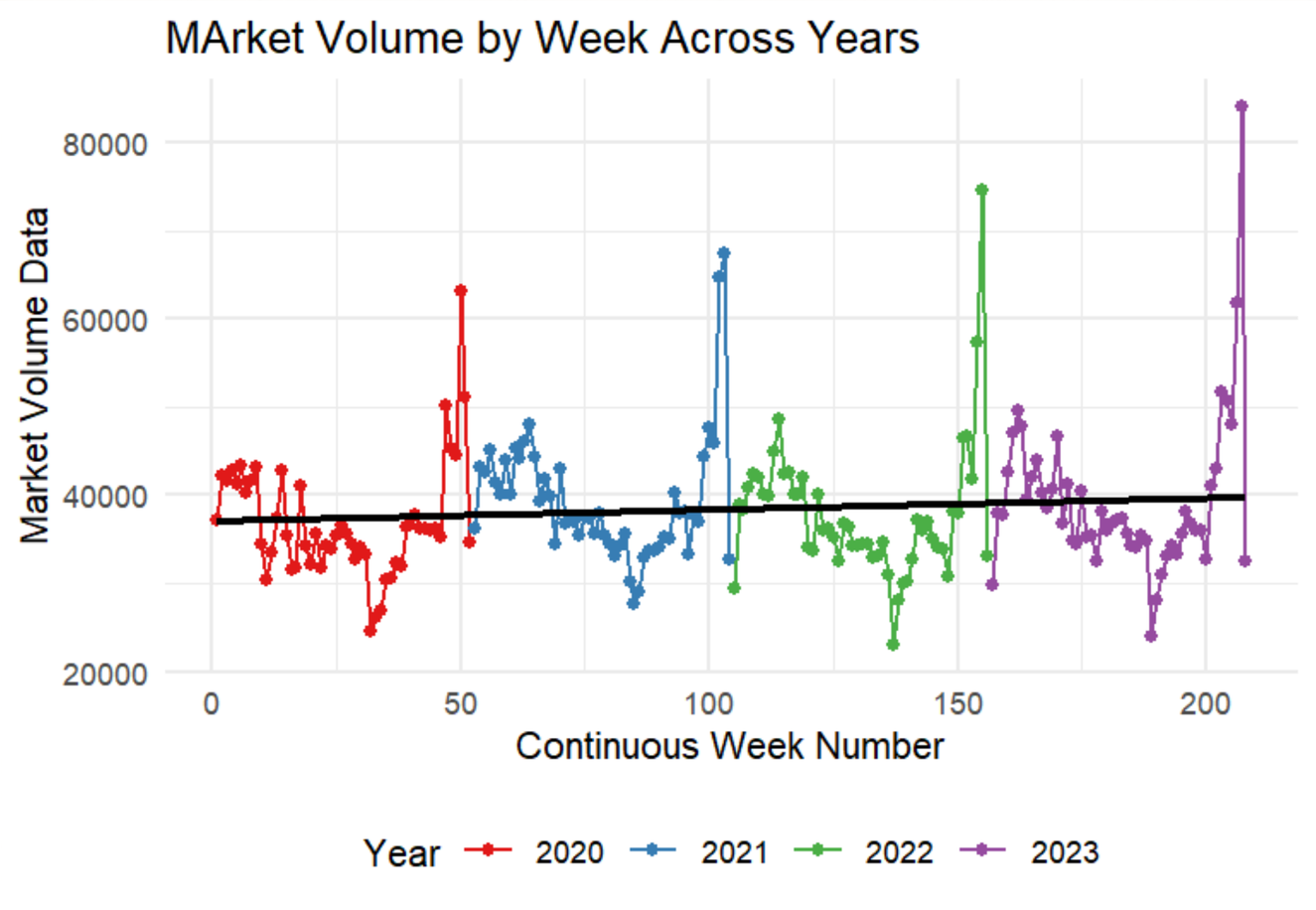
- In order to reduce the total number of variables that the model needs to deal with we decided to combine a few of the columns together. The reasoning behind combining them were:
  a. The ‘Totem Display in store’ and ‘Totem Display next to cashier’ were combined into a single column of ‘Totems’. The difference in the effectiveness of this media, in terms of sub categories would be very minuscule compared to the other media channels, giving us outputs which comparatively would not make sense. Also combining them made the data in this category more evenly spread.

III. PRELIMINARY DATA ANALYSIS

b. In the dataset, two main competitors were mentioned, “Mass” (Comp1) and “Pharma”(Comp2). This data was highly unsymmetric in terms of its distribution and hence showed high skewness. So we decided to opt for a median GRP value for each competitor for the week. Then the adstock was calculated on the basis of these median values.

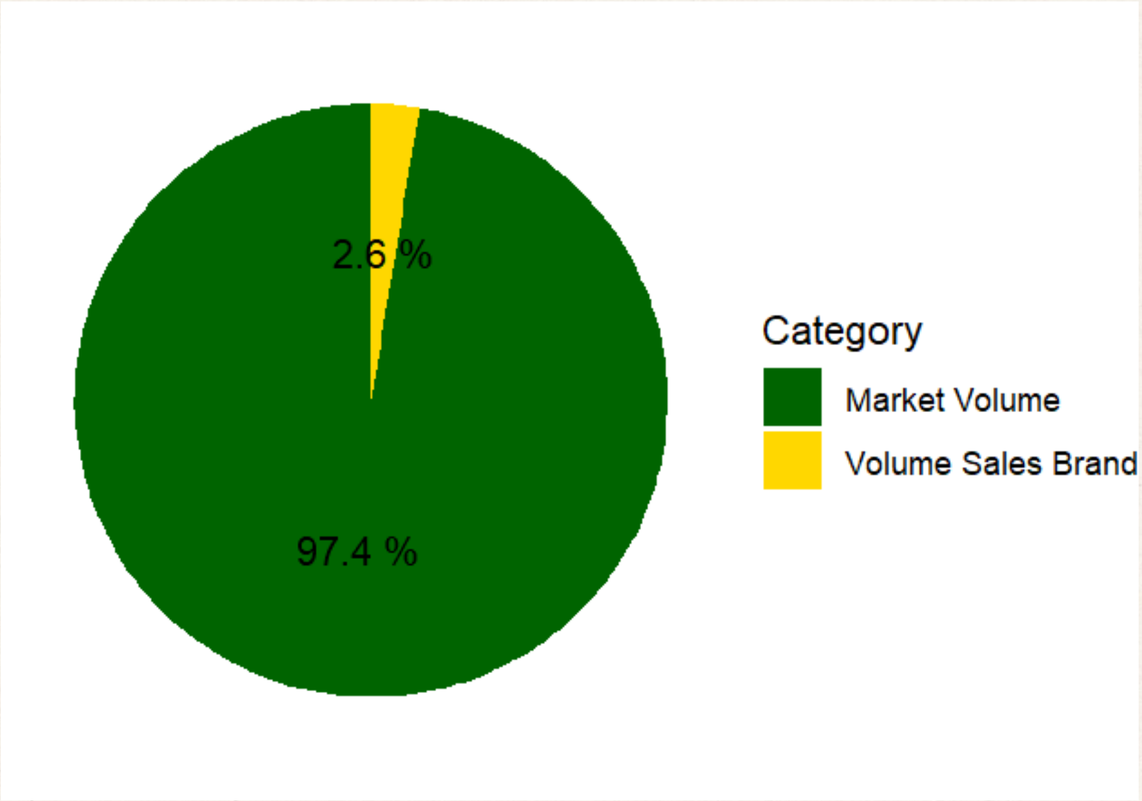
3. <u>Understanding the Market dynamics</u>

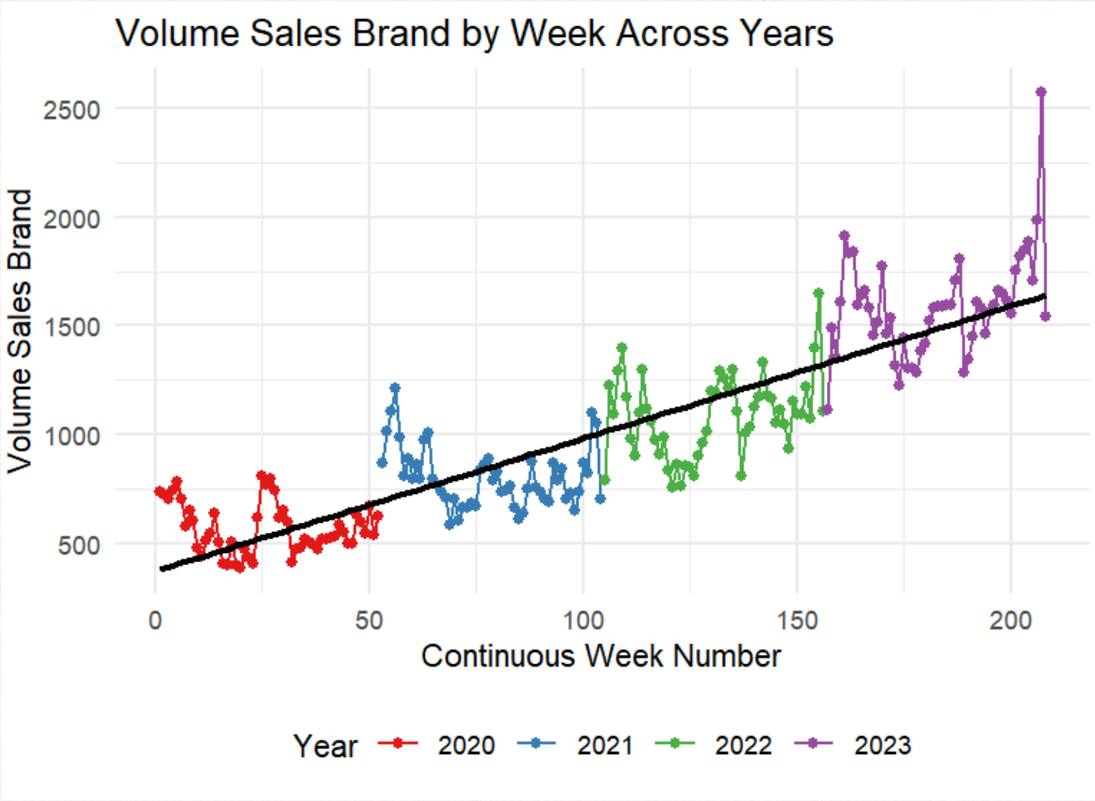
- For this section we are have provided an elaborate insight in the Manegerial Perspective slide. For the technical context, we first check the trend of the market, and compared it with the trend of our brand.
- We checked the trend of the market and the brand in terms of the amount of volume that was being sold.
- Based on this graph we can understand that the volume sales in this market are not growing, but they are not declining either. The trend also shows that the product sells a lot in during certain times a year, indicating strong seasonality of sales.



III. PRELIMINARY DATA ANALYSIS

- For our brand, we can clearly see that its volume sales are growing constantly through the years. This means that the brand is performing really well and is getting recognized by a larger group of people. It's current market share of 2.6% also shows that it has a great oppurtunity to expand. Based on these two results, we concluded that the current MMM structure present, works, but we would have to optimise it, to increase the return on investments (ROI) and the return on ad spend(ROAS).





IV. MODEL SELECTION, MODEL FIT, AND HYPERPARAMETER OPTIMIZATION

To create the Marketing Mix model for this brand we chose to create our model using Robyn. Robyn is a Semi-Automated Marketing Mix Modeling (MMM) aiming to reduce human bias by means of ridge regression and evolutionary algorithms.
In order to create a MMM model using Robyn, we first have to define the dependent and the independent variables. We chose our dependent variable to be *VALUE SALES BRAND*. We defined our independent variables in the following way,

INDEPENDENT VARIABLES:

- *PROMOTIONAL VARIABLES*: Value Sales in Promo price cut, Value Sales in Promo Bundle, Price index Vs Rest Market.
- *ONLINE MEDIA INVESTMENTS*: Youtube, Social Meta, Othe Video Publishers, Google Search, Amazon Search, Tiktok, Influencer Marketing Instagram and Tiktok.
- *OFFLINE MEDIA INVESTMENTS*: TV, Print, Free Samples Distribution
- *ONLINE MEDIA IMPRESSIONS*: Gross Rating Points, Youtube, Social Meta, Othe Video Publishers, Online Reviews, Tiktok, Influencer Marketing Instagram and Tiktok.

Additionally to create the model using Robyn, we need to define Robyn specific variables as well. These are:

- Paid Media Variables(paid_media_vars): These are the media variables with explicit and specific Investments. Data Transformation techniques will be applied to these variables in order to calculate the adstock effects.
- Paid Media Spends(paid_media_spends): These are the investments made for different media channels.

IV. MODEL SELECTION, MODEL FIT, AND HYPERPARAMETER OPTIMIZATION

- Organic Media Variables(organic_vars): These are the media variables without an explicit investment.
- Context Variables (context_vars) : These are variables that are not paid or Organic Media Variables. These are the variables which define the context in which the brand is working. This category includes things like competitor activity, seasonality, extra-ordinary events like Lockdowns,etc.

We have seperately provided the script for our code for Robyn. Henceforth, we are just explaining our perspectives behind our code and the selection of the optimum model.

- Hyperparameter Selection: Firstly, Robyn selects random values for the hyperparameters (theta, gamma, and alpha ranges) from a predefined space informed by business insights. These hyperparameters are crucial as they determine the effectiveness and longevity of advertising impacts.
- Adstock Model:The adstock model for each media channel is calculated to capture the diminishing returns of advertising over time. Three adstock models are available: Geometric, Weibull PDF, and Weibull CDF. For this report, the Geometric adstock transformation has been adopted.
- Mathematically, the Geometric adstock transformation is represented as: $M_t' = M_t + \theta M_{t-1}'$

  Where: M_t',M_t, and M_t' is the transformed media variable at time t. MtM_tMt is the original media variable at time t.

  θ(adstock rate) is the hyperparameter controlling the carryover effect, $0 \le \theta \le 10$. The half-life h is related to θ by the equation $\theta = 0.5^h$

IV. MODEL SELECTION, MODEL FIT, AND HYPERPARAMETER OPTIMIZATION

The performance metric of the model:

1. NRMSE (Normalized Root Mean Square Error): This metric measures the prediction error.
2. DECOMP.RSSD (Decomposition Root Sum of Squared Distance): This is a key metric invented by Robyn, representing the difference between the share of spend and the share of effect for paid media variables.

The objective function f maps hyperparameters θh to the performance metric: $f(\theta,h)$=Performance Metric.

<u>Bayesian Optimization:</u>

Bayesian optimization is used to find the optimal hyperparameters. A Gaussian Process (GP) models the objective function as a distribution over functions. By minimizing this function, which maps NRMSE and DECOMP.RSSD through an acquisition function, the next set of hyperparameters is determined. This iterative process continues until convergence, identifying the optimal set of adstock hyperparameters that represent the minimum curve point within the given space.

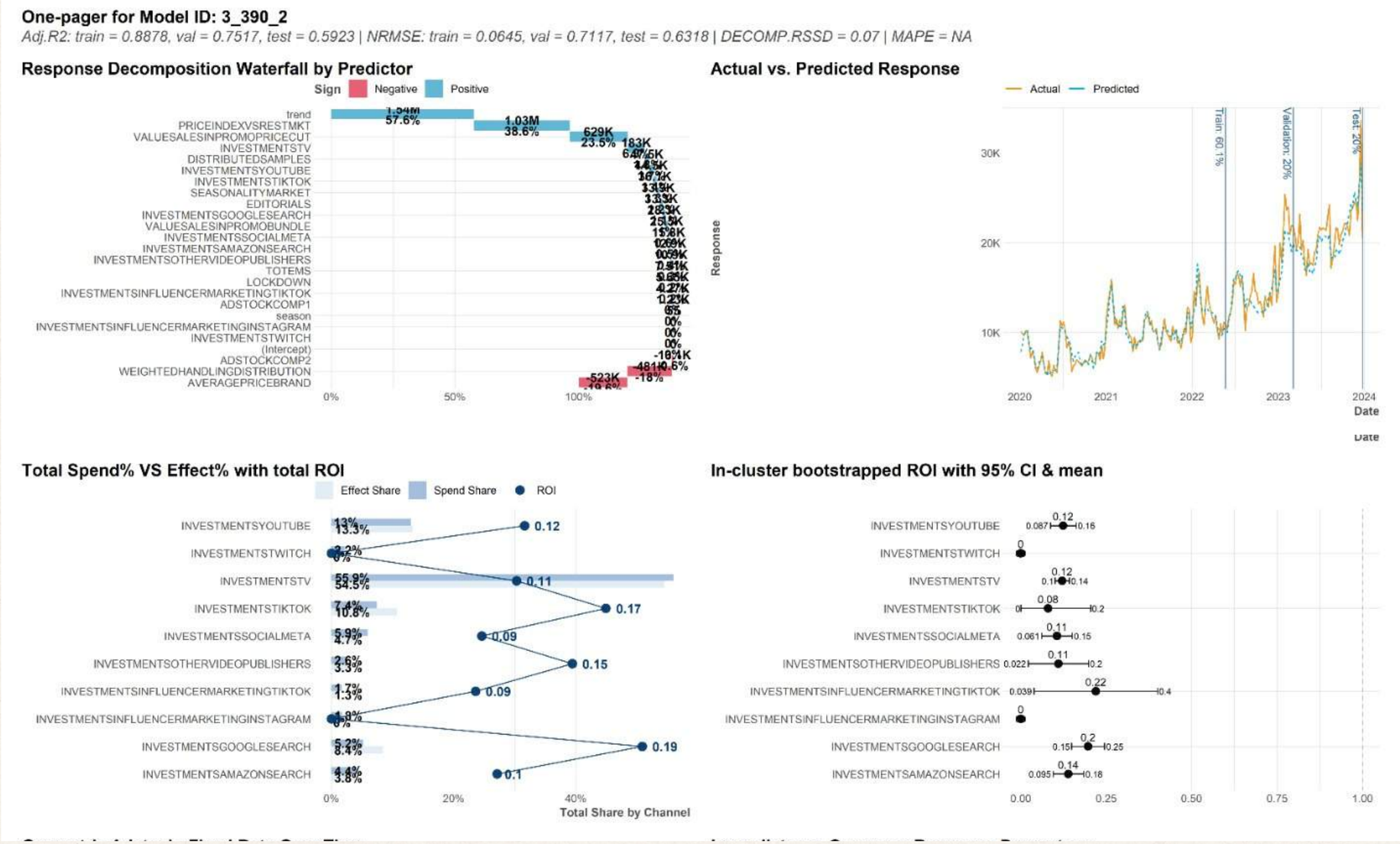
<u>Simultaneous Optimization:</u>

This optimization process runs concurrently with optimizing the set of Ridge regression coefficients. After transforming all media spends with the temporal set of hyperparameters iteratively, the model converges. The result is a set of hyperparameters that minimize the overall performance metrics, leading to an effective and efficient marketing mix model.

VI. MODEL EVALUATON AND COSIDERATIONS

- After running 5000 iterations, Robyn generates a cluster of top models based on NRMSE (Normalized Root Mean Square Error) and DECOMP.RSSD (Decomposition Root Sum of Squared Distance). From these clusters, the best models are chosen using business knowledge to evaluate the contribution of each media investment, ensuring they either positively impact sales or are close to neutral, while also considering the relevance of other covariates. Additionally, the evaluation includes analyzing the residual plot against fitted values to determine if the predicted sales values accurately reflect the underlying patterns of the real phenomenon and resemble white noise.
- Since the introduction of interaction terms to be modeled, direct solutions to incorporate media synergies into the model are not offered by Robyn. Therefore, they were manually added by multiplying each media investment covariate by the other media investments. Additionally, ranges for the carryover coefficients were defined based on geometric adstock. As a result, regression coefficients for each interaction term were obtained, and their carryover hyperparameters were optimized. This approach led to adjustments in the ROI values calculated by Robyn, as not making these adjustments would result in the underestimation of the actual ROI. For example, YouTube ROI values from Robyn do not take into account sales driven by the interaction between YouTube and TV spends. Results for Direct Effect ROI and Conditional ROI TotalI are shown on MODEL 1 ROI MODIFIED CALCULATIONS. Conditional ROI means ROI value under the condition there are investments on both each pair of media together.

SELECTED MODEL



SELECTED MODEL

VII. INTERPRETATION OF THE MODEL AND RESULTS ANALYSIS

Model 1 is considered the best model, revealing several key insights along with suggestions for each:

1. **Promo Strategies:**
   - **Insight:** More than 17% of the inflation-adjusted sales value is driven by the covariate VALUESALES IN PROMO, indicating that promotional strategies were effective throughout the modeling period.
   - **Suggestion**: Continue to invest in promotional activities and consider expanding them. Analyze which specific promotions are most effective and replicate or enhance these strategies.
2. **Marketing Synergies:**
   - **Insight:** The interaction between TV investment and YouTube has been highly effective, contributing approximately 10% to the sales value.
   - **Suggestion:** Maintain and possibly increase the coordinated marketing efforts between TV and YouTube. to further amplify marketing effectiveness.
3. **Industry Trends:**
   - **Insight:** The trend covariate, which is a function of time, highlights the growing opportunities within the industry.
   - **Suggestion:** Capitalize on the growing industry opportunities by staying ahead of market trends. Invest in market research to identify emerging opportunities and adapt strategies accordingly.
4. **Competitive Influence:**
   - **Insight:** Competitors' marketing activities have a negative impact on brand sales, emphasizing the need for strategic countermeasures.
   - **Suggestion:** Conduct a competitive analysis to understand competitors' strategies. Develop counter-marketing strategies to mitigate their impact, such as differentiated marketing messages or targeted promotions.

VII. INTERPRETATION OF THE MODEL AND RESULTS ANALYSIS

5. Distribution Strategies:

Insight: Weighted Handling Distribution (WEIGHTEDHANDLINGDISTRIBUTION) significantly impacts sales, suggesting that focusing on effective distribution strategies can maximize sales.

Suggestion: Optimize distribution channels to ensure product availability and visibility. Implement effective inventory management and logistics strategies to enhance distribution efficiency**.**

6.Promo Bundles:

Insight: VALUESALESINPROMOBUNDLE is 0, indicating that promotional bundles do not significantly impact sales.

Suggestion: Allocate resources to other promotional activities that yield better results, such as price cuts and advertising. Experiment with different promotional strategies and measure their effectiveness to determine the best approach.

7. Other Video Publishers:

Insight: INVESTMENTSOTHERVIDEOPUBLISHERS suggesting that investments in other video publishers do not significantly impact sales.

Suggestion: Reevaluate the effectiveness of these investments. Consider reallocating budget from other video publishers to more impactful channels like TV or YouTube, or experiment with diffe rent content or formats to see if performance improves.

VII. INTERPRETATION OF THE MODEL AND RESULTS ANALYSIS

8.Influencer Marketing on Instagram:

Insight: INVESTMENTSINFLUENCERMARKETINGINSTAGRAM is 0, indicating that influencer marketing on Instagram does not significantly impact sales.

Suggestion: Assess the strategies used in influencer marketing on Instagram. Investigate whether the right influencers are being targeted, the content strategy is effective, or if other platforms might offer better engagement. Reallocate resources to more effective marketing channels if necessary

9.**TV Investment:**

Insight: TVINVESTMENT has low ROI.

Suggestion: Evaluate the effectiveness of TV investments by analyzing the ROI metrics more closely. Consider adjusting the timing, frequency, or content of TV ads to improve effectiveness.

10.Amazon Search Investments:

Insight: INVESTMENTSAMAZONSEARCH has the highest ROI.

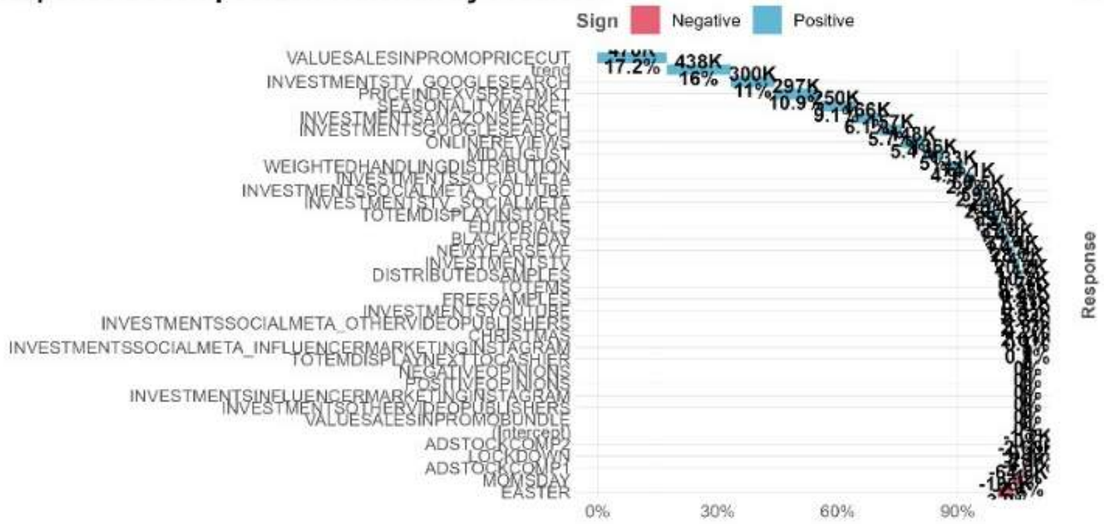
Suggestion: Allocate more resources to Amazon Search investments due to its high ROI. Explore opportunities to further enhance performance through keyword optimization, sponsored product ads, and product listing optimizations on Amazon's platform. Monitor trends and adjust strategies accordingly to maintain or improve ROI over time.

EXPERIMENTATIONS WITH OUR MODEL

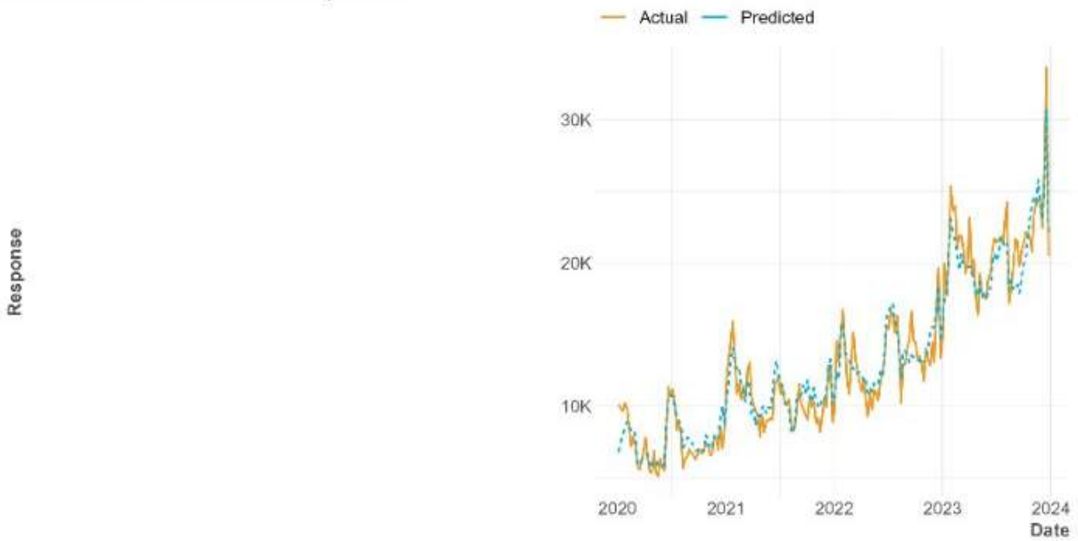
One-pager for Model ID: 5_636_1

Adj.R2: train = 0.9392 | NRMSE: train = 0.0433 | DECOMP.RSSD = 0.5908 | MAPE = NA

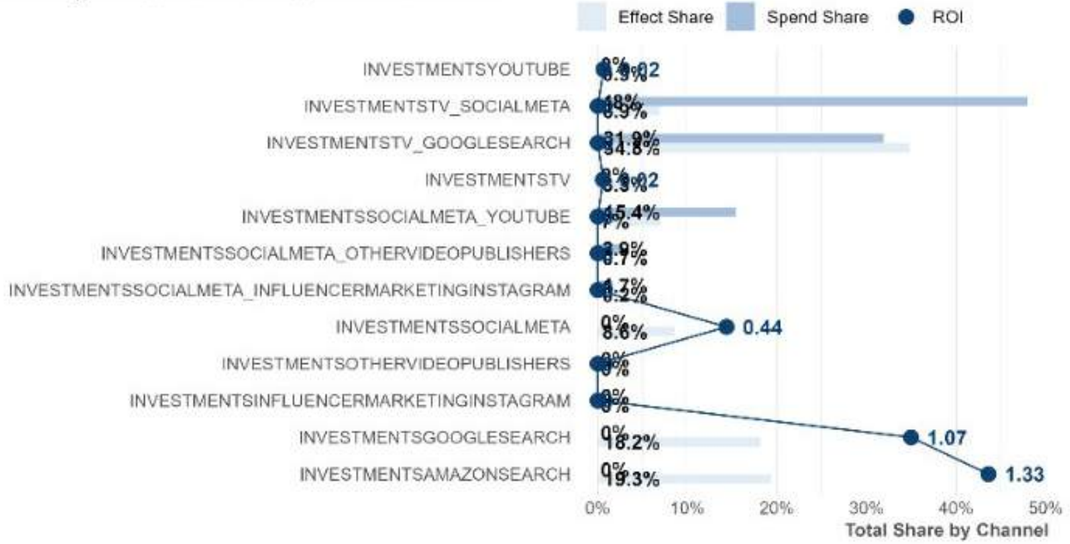
Response Decomposition Waterfall by Predictor



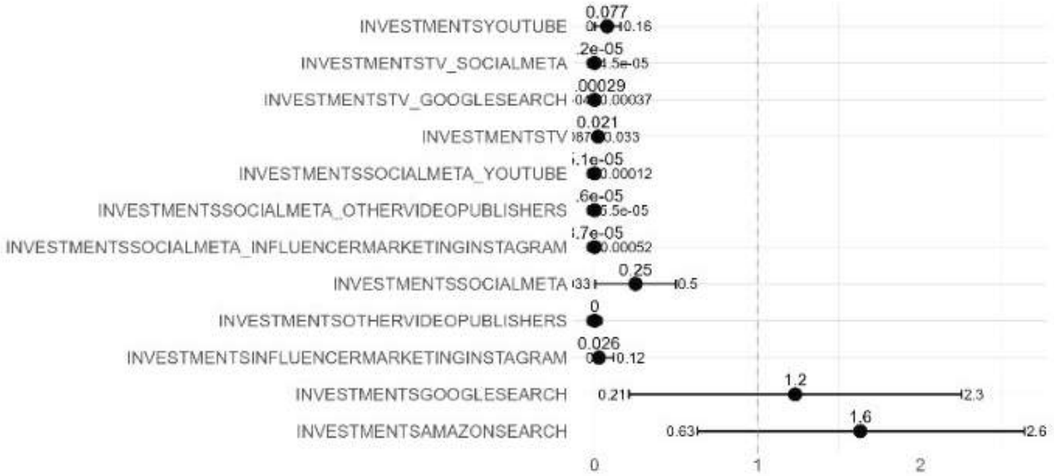
Actual vs. Predicted Response



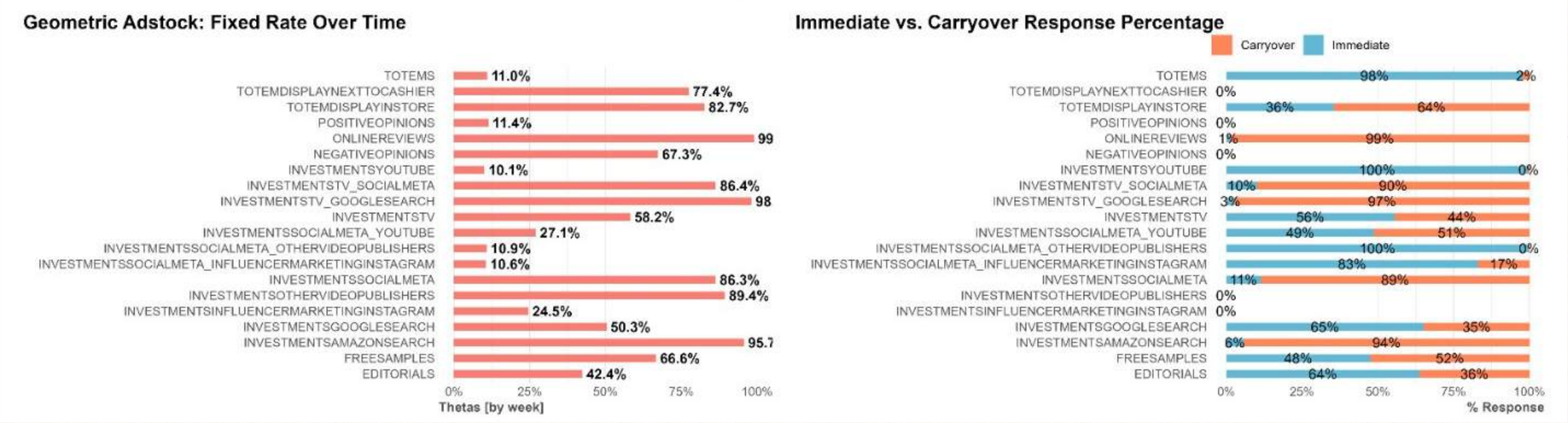
Total Spend% VS Effect% with total ROI

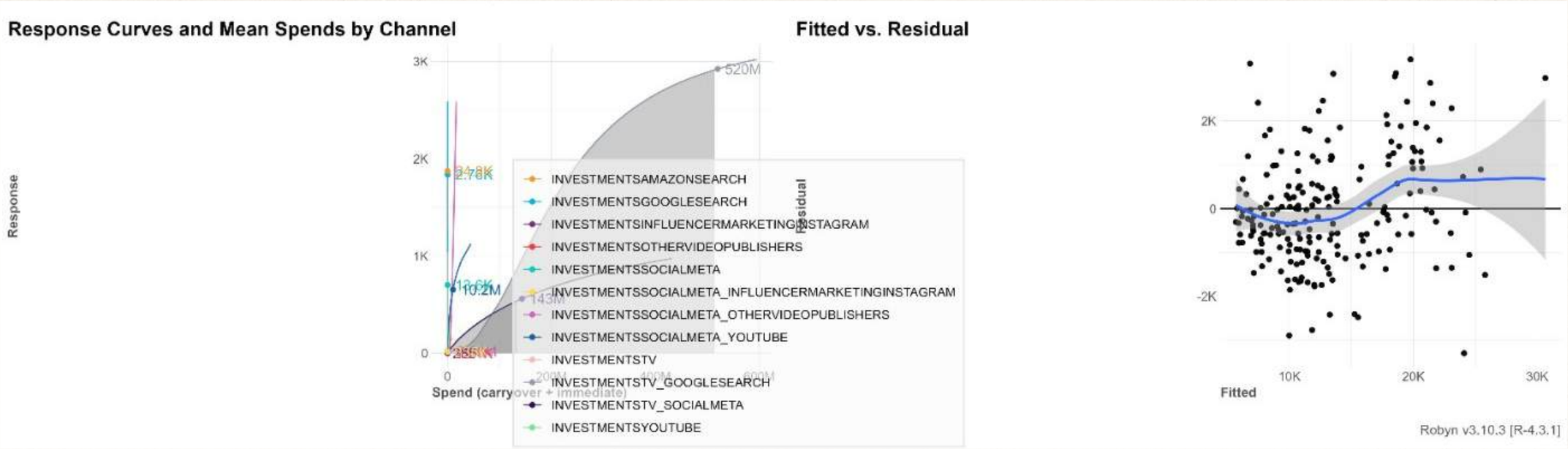


In-cluster bootstrapped ROI with 95% CI & mean



EXPERIMENTATIONS WITH OUR MODEL





MODEL 2

ROI MODIFIED CALCULATIONS

Channel	Effect Share	Total spend	ROI Direct Effect	Indirect ROI	Conditional ROI Total
INVESTMENTSTV	0.03	1592573	0.02	0	0.57
INVESTMENTSYOUTUBE	0.01	371571.39	0.02	0	0.37
INVESTMENTSOTHERVIDEOPUBLISHERS	0.00	73047.51	0	0	0.08
INVESTMENTSSOCIALMETA	0.09	168931.79	0.44	0	1.01
INVESTMENTSGOOGLESEARCH	0.18	146798.67	1.07	0	1.62
INVESTMENTSINFLUENCERMARKETINGINSTAGRAM	0.00	50215.669	0	0	0.03
INVESTMENTSTV_GOOGLESEARCH	0.35		0	0.55	na
INVESTMENTSTV_SOCIALMETA	0.07		0	0.11	na
INVESTMENTSSOCIALMETA_YOUTUBE	0.07		0	0.35	na
INVESTMENTSSOCIALMETA_OTHERVIDEOPUBLISHERS	0.01		0	0.08	na
INVESTMENTSSOCIALMETA_INFLUENCERMARKETINGINSTAGRAM	0.00		0	0.03	na
INVESTMENTSAMAZONSEARCH	0.19	125138.76	1.33	0	1.33

FORMULA:

ON ROBYN ROI is calculated by dividing the incremental contribution by the media spend. Mathematically, it can be expressed as:

$$\text{ROI}_{\text{channel}} = \frac{\text{Incremental Contribution}_{\text{channel}}}{\text{Media Spend}_{\text{channel}}}$$

Where:

Incremental contrubution: The contribution of each media channel is calculated by multiplying the media spend by the corresponding regression coefficient. This gives the estimated incremental contribution to the target variable due to each media channel.